

AIMOTOR with CANOPEN driver manual

A: Product introduction

1.1 Product Overview

AIMOTOR driver uses CANOPEN bus communication interface, CANOPEN slave station technology, vector control technology, built-in micro-segmentation technology, adaptive filtering technology, closed-loop control technology integration, to achieve real-time control and real-time data transmission of the system, optimize the performance of the motor: has excellent stability and ultra-low noise; Smooth and accurate pure sinusoidal current vector control technology effectively reduces motor heating.

1.2 Product Characteristics

- The new generation of 32-bit ARM technology, high cost performance, good stability, noise, vibration performance is superior
- CANOPEN slave technology, support CIA301 and CIA402 sub-protocols, support CSP, PV, PP, PT, HM mode
- Users can set system parameters through CANOPEN bus or serial communication.

1.3 Networking scheme

The transmission distance of CAN bus is directly related to the baud rate and communication cable, and the relationship between the maximum bus line length and the baud rate is shown in the following table:

The default NodeId=1, rate =500kbps, if you want to change, please use AIMotor series RS485 debugging PC to change H0C_00 and H0C-08 parameters;

transmission distance	BAUD	Number of Nodes	Twisted pair cable diameter
25m	1Mbps	64	0.2mm ²
95m	500kbps	64	0.3mm ²
560m	100kbps	64	0.5mm ²
1100m	50kbps	64	0.75mm ²

Figure 1 Networking scheme 1

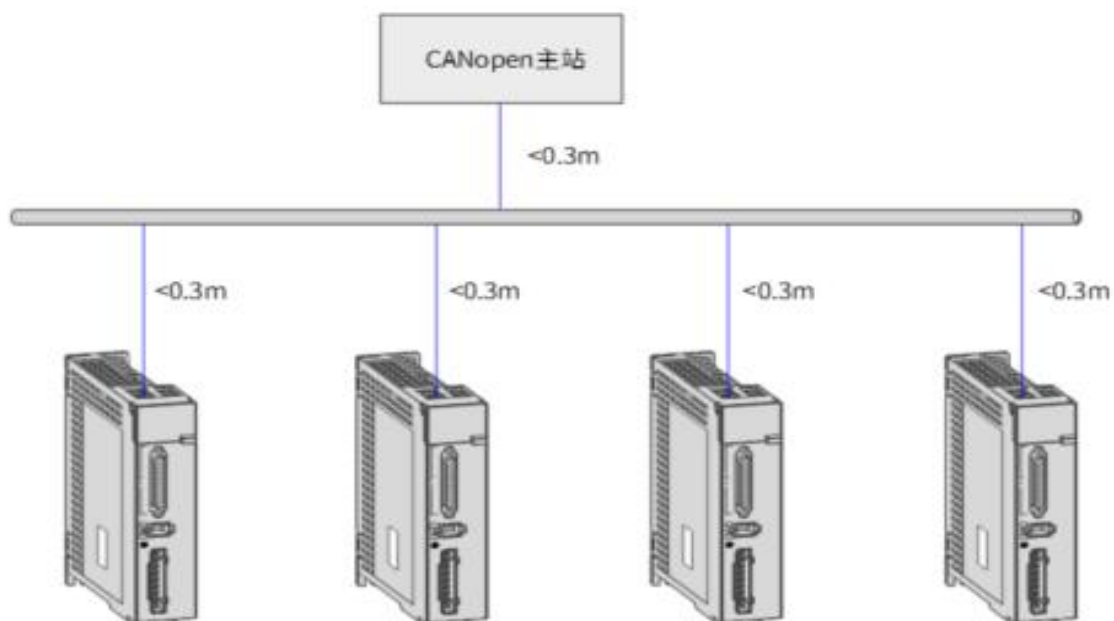
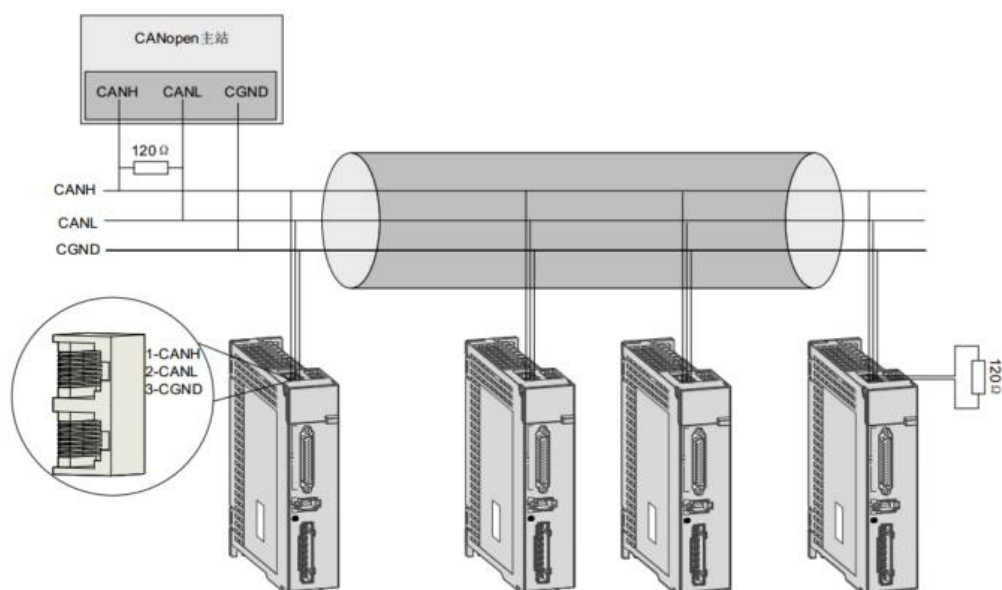


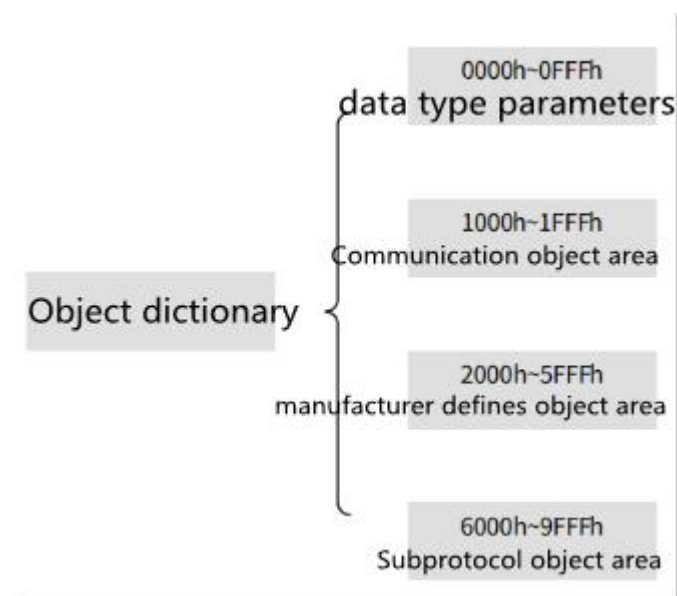
Figure 2 Networking scheme 2



B: Parameter description and setting of object dictionary

2.1 Object dictionary

The object dictionary is the most important part of the device specification. It is an ordered set of parameters and variables, containing all parameters of the device description and the network state of the device. A set of objects that can be accessed through a network in an ordered, predefined manner. CANopen protocol uses an object dictionary with a 16-bit index and an 8-subindex. The structure of the object dictionary is shown in the figure as below:



0000h-0FFFh User has no access significance.

1000h-1FFFh defines communication object parameters and device types according to CIA301 protocol.

2000h-5FFFh Manufacturer defined object area, we define all AIMOTOR drive parameter function code objects in this area;

6000h-9FFFh Some common CIA402 control objects defined by the CIA402 protocol

The corresponding relationship between AIMOTOR drive parameter function code and object dictionary is as follows: Object dictionary index = 0x2000 + function code group number; Object dictionary subindex = hexadecimal + 1 offset in function code set

Example of mapping relationship between drive parameter function code and object dictionary:

Parameter function code			Corresponding object dictionary address	
Parameter function code name	Function code group number (hexadecimal)	In-group bias	Index (hexadecimal)	Subindex (hexadecimal)
H02-00	0x02	00	0x2002	0x01
H03-09	0x03	09	0x2003	0x0A
H0B-26	0x0B	26	0x200B	0x1B
H12-20	0x12	20	0x2012	0x15

2.1.1 CIA301 subprotocol communication parameters

Index	Sub-Index	Name	Description	Type	Nature	False	Range
1000	0	device type	Consistent with CIA rules	Uint32	RO	-	-
1001	0	Error register	No effect, reserved to increase	Uint8	RO	0	-
1005	0	Sync Messages COB-ID	Used in the periodic synchronization function	Uint32	RW	0x80	0~0xFFFFFFFF
1006	0	Synchronization cycle		Uint32	RW	0	0~0xFFFFFFFF
1008	0	Device name	Screen printing is the main product	str	RO		0~32767
1009	0	Hardware version	PCB Version	str	RO	-	-
100A	0	Software version	Burning program version	str	RO	-	-
100C	0	Node guard time	Used in the node daemon function	Uint16	RW	0	0~32767
100D	0	Life factor		Uint8	RW	0	0~255
1010	00	Subindex number	No effect, reserved to increase. To save the parameter to eeporm, operate H0C_13	Uint8	RO	4	-
	01	Save all parameters		Uint32	RW	0	0~0xFFFFFFFF
	02	Save communication parameters		Uint32	RW	0	0~0xFFFFFFFF
	03	Save motion parameters		Uint32	RW	0	0~0xFFFFFFFF
	04	Save manufacturer's parameters		Uint32	RW	0	0~0xFFFFFFFF
1014	0	Emergency message COB-ID	Emergency packet identifier	Uint32	RW	0x80	0~0xFFFFFFFF
1016	0	Subindex number	Set the consumer heartbeat time	Uint8	RO	1	-
	1	Consumer heartbeat time		Uint32	RW	0	0~0xFFFFFFFF
1017	0	Producer heartbeat time	Set the heartbeat time of the producer	Uint32	RW	0	0~0xFFFFFFFF
1029	0	Subindex number	Error behavior object	Uint8	RO	1	-
	1	Communication Error		Uint8	RW	0	0~255
1200	0	Subindex number	Setup SDO Communication COB-ID	Uint8	RO	2	-
	1	Client to server COB-ID		Uint32	RW	0x600+ NodeID	0~0xFFFFFFFF

	2	Server to client COB-ID		Uint32	RW	0x580+ NodeID	0~0xFFFFFFFF
Index	Sub-Index	Name	Description	Type	Nature	False	Range
1400	0	Subindex number	Setup RPDO1 communication interface	Uint8	RO	6	-
	1	COB-ID of RPDO1		Uint32	RW	0x200	0~0xFFFFFFFF
	2	Transmission type of RPDO1		Uint8	RW	1	0~255
1401	0	Subindex number	Setup RPDO2 communication interface	Uint8	RO	6	-
	1	COB-ID of RPDO2		Uint32	RW	0x300	0~0xFFFFFFFF
	2	Transmission type of RPDO2		Uint8	RW	1	0~255
1402	0	Subindex number	Setup RPDO3 communication interface	Uint8	RO	6	-
	1	COB-ID of RPDO3		Uint32	RW	0x400	0~0xFFFFFFFF
	2	Transmission type of RPDO3		Uint8	RW	1	0~255
1403	0	Subindex number	Setup RPDO4 communication interface	Uint8	RO	6	-
	1	COB-ID of RPDO4		Uint32	RW	0x500	0~0xFFFFFFFF
	2	Transmission type of RPDO4		Uint8	RW	1	0~255
1600	0	Number of RPDO1 mappings	Setup RPDO Group 1 Mapping object ;	Uint16	RW	3	0~32767
	1	RPDO1 Mapping object 1	Factory default mapping is as follows:	Uint16	RW	0x60400010	0~65535
	2	RPDO1 Mapping object 2	Subindex 1 maps 6040 control words;	Int8	RW	0x60600008	0~128
	3	RPDO1 Mapping object 3	Subindex 2 maps 6060 control mode;	Uint32	RW	0x607A0020	0~0xFFFFFFFF
	4-8	RPDO1 Mapping object 4~8	Subindex 3 maps 607A target location; Subindexes 4-8 are unmapped and can be allocated on demand.	Uint32	RW	0x00000000	0~0xFFFFFFFF
1601	0	Number of RPDO2 mappings	Set an RPDO group 2 mapping object.	Uint16	RW	2	0~32767
	1	RPDO2 Mapping object 1	The factory default mapping is as follows:	Uint32	RW	60FF0020	0~0xFFFFFFFF
	2	RPDO2 Mapping object	Subindex 1 maps 60FF target	Int16	RW	60710010	0~32767

		2	speed;				
	3-8	RPDO2 Mapping object 3~8	Subindex 2 maps 6071 target torque; Subindexes 3-8 are unmapped and can be allocated on demand.	Uint32	RW	0x00000000	0~0xFFFFFFFF
1602	00	Number of RPDO3 mappings	Set the RPDO group 3 mapping object.	Uint16	RW	0	0~32767
	01-08	RXPDO3 Maps objects 1 to 8	Subindexes 1-8 are unmapped and can be allocated on demand.	Uint32	RW	0x00000000	0~0xFFFFFFFF
1603	0	Number of RPDO4 mappings	Set an RPDO group 4 mapping object.	Uint16	RW	0	0~32767
	01-08	RXPDO4 Maps objects 1 to 8	Subindexes 1-8 are unmapped and can be allocated on demand.	Uint32	RW	0x00000000	0~0xFFFFFFFF

Index	Sub-Index	Name	Description	Type	Nature	False	Range
1800	0	Subindex number	Set TPDO1 communication parameters	Uint8	RO	6	-
	1	COB-ID of TPDO1		Uint32	RW	0x180	0~0xFFFFFFFF
	2	Transport type of TPDO1		Uint8	RW	1	0~255
1801	0	Subindex number	Set TPDO2 communication parameters	Uint8	RO	6	-
	1	COB-ID of TPDO2		Uint32	RW	0x280	0~0xFFFFFFFF
	2	Transport type of TPDO2		Uint8	RW	1	0~255
1802	0	Subindex number	Set TPDO3 communication parameters	Uint8	RO	6	-
	1	COB-ID of TPDO3		Uint32	RW	0x380	0~0xFFFFFFFF
	2	Transport type of TPDO3		Uint8	RW	1	0~255
1803	0	Subindex number	Set TPDO4 communication parameters	Uint8	RO	6	-
	1	COB-ID of TPDO4		Uint32	RW	0x480	0~0xFFFFFFFF
	2	Transport type of TPDO4		Uint8	RW	1	0~255
1A00	0	Number of TPDO1 mappings	Set the TPDO group 1 mapping object.	Uint16	RW	3	0~32767

	1	TPDO1 Mapping object 1	The factory default mapping is as follows: Subindex 1 maps 6041 status words; Subindex 2 maps 6061 display mode; Subindex 3 maps 6064 actual location; Subindexes 4-8 are unmapped and can be allocated on demand.	Uint16	RW	0x60410010	0~65535
	2	TPDO1 Mapping object 2		Int8	RW	0x60610008	0~128
	3	TPDO1 Mapping object 3		Uint32	RW	0x60640020	0~0xFFFFFFFF
	4-8	TPDO1 Mapping object 4~8		Uint32	RW	0x00000000	0~0xFFFFFFFF
1A01	0	Number of TPDO2 mappings	Set the mapping object of TPDO group 2.	Uint16	RW	2	0~32767
	1	TPDO2 Mapping object 1	The factory default mapping is as follows: Subindex 1 maps 606C actual speed; Subindex 2 maps 6077 actual torque; Subindexes 3-8 are unmapped and can be allocated on demand.	Uint32	RW	606C0020	0~0xFFFFFFFF
	2	TPDO2 Mapping object 2		Int16	RW	60770010	0~32767
	3-8	TPDO2 Mapping object 3~8		Uint32	RW	0x00000000	0~0xFFFFFFFF
1A02	00	Number of RPDO3 mappings	Set the mapping object of TPDO group 3.	Uint16	RW	0	0~32767
	01-08	RXPDO3 Maps objects 1 to 8	Subindexes 1-8 are unmapped and can be allocated on demand.	Uint32	RW	0x00000000	0~0xFFFFFFFF
1A03	0	Number of RPDO4 mappings	Set the mapping object of TPDO group 4.	Uint16	RW	0	0~32767
	01-08	RXPDO4 Maps objects 1 to 8	Subindexes 1-8 are unmapped and can be allocated on demand.	Uint32	RW	0x00000000	0~0xFFFFFFFF

2.1.2 Manufacturer custom parameters

2000h-5FFFh Manufacturer defined object area, we define all AIMOTOR drive parameter function code objects in this area;

Each parameter has a corresponding dictionary object index and sub-index, AIMOTOR drive parameter function code and object dictionary corresponding relationship is as follows:

Object dictionary index = 0x2000 + function code group number; Object dictionary subindex = hexadecimal + 1 offset within function code set;

Example: Mapping relationship between drive parameter function code and object dictionary

Parameter function code			Corresponding object dictionary address	
Parameter function code name	Function code group number (hexadecimal)	In-group bias	Index (hexadecimal)	Subindex (hexadecimal)
H02-00	0x02	00	0x2002	0x01
H03-09	0x03	09	0x2003	0x0A
H0B-26	0x0B	26	0x200B	0x1B
H12-20	0x12	20	0x2012	0x15

Due to the excessive parameter function code, it is not listed in the CANOPEN manual, please refer to the same series of standard pulse product manual parameter description.

◆ When using SDO communication to read and write the manufacturer's custom parameter object, please pay attention to the attribute of the parameter;

1.If the modify mode property is < Enable disconnect > parameter, you need to disconnect the motor enable and then write the parameter object value;

2.If the factory value attribute is < registered value > parameter, the user can only read but not write;

◆ H00 group, H01 group, H0B group parameters, users can only read but not write;

◆ SDO communication error abort code

Abort code	Tag	Description
0x00000000	OD_SUCCESSFUL	SDO communication successful
0x06010001	OD_READ_NOT_ALLOWED	Trying to read only write objects
0x06010002	OD_WRITE_NOT_ALLOWED	Attempting to write to a read-only object
0x06020000	OD_NO_SUCH_OBJECT	Object does not exist in the dictionary
0x06040041	OD_NOT_MAPPABLE	Objects cannot be mapped in PDO
0x06060000	OD_ACCES_FAILED	Access failed due to hardware error
0x06070010	OD_LENGTH_DATA_INVALID	Data type and length of the service parameter do not match
0x06090011	OD_NO_SUCH_SUBINDEX	Word index does not exist
0x06090030	OD_VALUE_RANGE_EXCEEDED	Out of parameter value range (write access only)
0x06090031	OD_VALUE_TOO_LOW	Parameter value is too high
0x06090032	OD_VALUE_TOO_HIGH	Parameter value is too low
0x05030000	SDOABT_TOGGLE_NOT_ALTERNED	Switching bits do not alternate
0x05040000	SDOABT_TIMED_OUT	SDO protocol communication timed out
0x05040001	SDOABT_CS_NOT_VALID	Command specifier is invalid
0x05040002	SDOABT_INVALID_BLOCK_SIZE	Invalid block size (block mode only)
0x05040005	SDOABT_OUT_OF_MEMORY	Memory Overflow
0x08000000	SDOABT_GENERAL_ERROR	SDO message size is incorrect
0x08000021	SDOABT_LOCAL_CTRL_ERROR	Due to local control, data cannot be transferred or stored in the application

2.1.3 Cia402 shaft control parameter group

Index	Sub-Index	Name	Description	Type	Nature	False	Range
6040	0	Control Word	For details, see section "2.4 6040 Control Words".	Uint16	RW	0	0~65535
6041	0	Status Word	For details, see section "2.5 6041 Status Word".	Uint16	RO	-	0~65535
605A	0	Quick stop code	0: free stop, free state 1: Stop at the deceleration set by 6084h 2: Stop at the speed set at 6085h	Uint16	RW	1	0~65535
6060	0	Run mode setting	0: has no effect 1: PP(contour position mode) 3: PV(Contour speed mode) 4: PV(Contour torque mode) 6: HM(back to origin mode) 8: CSP(Cyclic synchronous position mode)	Int8	RW	8	0~255
6061	0	Run mode state	Query the status of 6060H. 0: has no effect 1: PP(Location mode) 3: PV(Speed mode) 4: PV(Contour torque mode) 6: HM(back to origin mode) 8: CSP(Cyclic synchronous position mode)	Int8	RO	-	0~255
60FC	0	Internal instruction position	Position instructions generated within the drive; Unit: Enc	Int32	RO	-	-2147483647~2147483647
6062	0	Internal instruction position	Position instructions generated within the drive; Unit: Pul	Int32	RO	-	-2147483647~2147483647
6063	0	Actual position	Actual motor location, Unit: Enc	Int32	RO	-	-2147483647~2147483647
6064	0	Actual position	Actual motor location, Unit: Pul	Int32	RO	--	-2147483647~2147483647
606B	0	Internal command speed	Speed instructions generated within the driver; Unit: Pul/s	Int32	RO	-	-2147483647~2147483647
606C	0	Actual speed	Actual motor speed, Unit: Pul/s	Int32	RO	-	-2147483647~2147483647

6071	0	Target torque	Set PT(Contour torque mode) running torque; (1000 indicates rated torque) Unit: 0.1%	Int16	RW	0	-3000~3000
6074	0	Internal instruction torque	Torque instructions generated inside the drive; Unit: 0.1%	Int16	RO	-	-3000~3000
6077	0	Actual torque	Actual torque of the motor, Unit: 0.1%	Int16	RO	-	-3000~3000
Index	Sub-Index	Name	Description	Type	Nature	False	Range
607A	0	Target location	Set PP(contour position mode) running position; Unit: Pul	Int32	RW	0	-2147483647~2147483647
607C	0	Origin Offset	Set HM(back to origin mode) origin offset; Unit: Pul	Int32	RW	0	-2147483647~2147483647
607E	0	Command polarity	Set the instruction direction logic in PP/PV/PT contour mode;	Uint8	RW	0	0~255
607F	0	Maximum profile velocity	Set the maximum speed limit for PT(contour torque mode) operation; Unit: Pul/s	Uin32	RW	0	-2147483647~2147483647
6081	0	Profile velocity	Set the maximum speed of PP(contour position mode); Unit: Pul/s	Int32	RW	0	-2147483647~2147483647
6083	0	Contour acceleration	Set the acceleration of PP(profile position mode) and PV(profile velocity mode); Unit: Pul/s^2	Int32	RW	100000	-2147483647~2147483647
6084	0	Contour deceleration	Set the speed reduction of PP(contour position mode) and PV(contour velocity mode); Unit: Pul/s^2	Int32	RW	100000	-2147483647~2147483647
6085	0	Quick stop and deceleration	Set quick stop and speed reduction in PP/PV/PT profile mode; Unit: Pul/s^2	Int32	RW	500000	-2147483647~2147483647
6098	0	Return to origin mode	Current can be set way back to the origin of value is: 01/02/04/06/17/18/24/28/33/34/35/37/38; For details, see section '6.5 Back to Origin Mode	Int8	RW	35	0~255

			Method'.				
6099	0	Subindex number	-	UInt8	RO	2	0~255
	1	Origin velocity 1	HM(Back to origin mode) Approach to the origin speed: high speed to find the origin; Unit: Pul/s	Int32	RW	10000	-2147483647~ 2147483647
	2	Origin velocity 2	HM(back to origin mode) to determine the origin speed: low speed to find the origin; Unit: Pul/s	Int32	RW	1000	-2147483647~ 2147483647
609A	0	Return to origin acceleration and deceleration	The rate of addition and subtraction of HM(return to origin mode) back to origin; Unit: Pul/s^2	Int32	RW	10000	-2147483647~ 2147483647
60FF	0	target speed	Set the PV(contour speed mode) running speed; Unit: Pul/s	Int32	RW	0	-2147483647~ 2147483647
6502	0	operation mode supported by the driver	Used to query the CIA402 motion control mode supported by the product	UInt32	RO	0x00AD	0~0xFFFFFFFF

2.2 Each control mode and associated object dictionary

Control Mode	Index + subindex	Name	Data type	read-write access permission	Unit	Required or not
Synchronous Position Mode (CSP)	6060	Control mode setting =8	Int8	RW	-	√
	6040	Control word	UInt16	RW	-	√
	607A	Target position	Int32	RW	Pul	√
	6061	Control mode feedback	Int8	RO	-	
	6041	Status word	UInt16	RO	-	√
	6064	Real position	Int32	RO	Pul	√
	606C	Real speed	Int32	RO	Pul/s	
	6077	Read torque	Int16	RO	0.1%	
Profile Position Mode (PP)	6060	Control mode setting =1	Int8	RW	-	√
	6040	Control word	UInt16	RW	-	√
	607A	Target position	Int32	RW	Pul	√
	6081	Maximum speed	Int32	RW	Pul/s	√
	6083	Acceleration	DINT	RW	Pul/s ²	√
	6084	Deceleration	DINT	RW	Pul/s ²	√
	6061	Control mode feedback	Int8	RO	-	
	6041	Status word	UInt16	RO	-	√
	6064	Real position	Int32	RO	Pul	
	606C	Real speed	Int32	RO	Pul/s	
	6077	Read torque	Int16	RO	0.1%	
Profile Velocity Mode (PV)	6060	Control mode setting =3	Int8	RW	-	√
	6040	Control word	UInt16	RW	-	√
	60FF	Target speed	DINT	RW	Pul/s	√
	6083	Acceleration	DINT	RW	Pul/s ²	√
	6084	Deceleration	DINT	RW	Pul/s ²	
	6061	Control mode	Int8	RO	-	

		feedback				
	6041	Status word	Uint16	RO	-	
	6064	Real position	Int32	RO	Pul	
	606C	Real speed	Int32	RO	Pul/s	
	6077	Read torque	Int16	RO	0.1%	
Contour torque mode (PT)	6060	Control mode setting =4	Int8	RW	-	√
	6040	Control word	Uint16	RW	-	√
	6071	Target torque	Int16	RW	0.1%	√
	607F	Maximum speed limit	Uint32	RW	Pul/s	√
	6061	Control mode feedback	Int8	RO	-	
	6041	Status word	Uint16	RO	-	
	6064	Real position	Int32	RO	Pul	
	606C	Real speed	Int32	RO	Pul/s	
	6077	Read torque	Int16	RO	0.1%	

Control Mode	Index + subindex	Name	Data type	read-write access permission	Unit	Required or not
Back to origin mode (HM)	6060	Control mode setting =3	Int8	RW	-	√
	6040	Control word	Uint16	RW	-	√
	6098	Return-to-zero method	Int8	RW	-	√
	6099:01	Search origin high speed 1	Int32	RW	Pul/s	√
	6099:02	Find origin low speed 2	Int32	RW	Pul/s	√
	609A	Origin acceleration	Int32	RW	Pul/s^2	√
	607C	Origin Offset	Int32	RW	Pul	
	6061	Control mode feedback	Int8	RO	-	
	6041	Status word	Uint16	RO	-	
	6064	Real position	Int32	RO	Pul	
	606C	Real speed	Int32	RO	Pul/s	
	6077	Read torque	Int16	RO	0.1%	
Other auxiliary associated parameters	605A	Quick stop code	Uint16	RW	-	
	6085	Crash stop to reduce speed	Uint32	RW	Pul/s^2	

	607E	Command polarity	Uint8	RW	-	
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No matter which control mode is used to control the slave station, it is inseparable from the read and write operation of the two object dictionaries of 6040H(control word) and 6041H(status word). The master and slave station use these two object dictionaries as transmission media to realize command delivery and status monitoring. The following sections focus on the definitions and meanings of the individual bits of the two object dictionaries.

2.3 Cia402 object dictionary position, speed, acceleration and deceleration unit calculation

The speed value in CIA402 protocol is not the unit of speed per minute r/min as we understand, but the unit of instruction pulse per second Pul/s; The unit of acceleration and deceleration is Pul/s²;

If I want to set the speed per minute, I should first convert minutes into seconds, and then multiply the number of pulses of the motor 1 turn;

AIMotor series drive factory default 1000 pulse motor rotation 1 turn; If you change the pulse number of 1 turn, please use debugging software or SDO communication mode to set H05-07 electronic gear molecule and H05-09 electronic gear denominator. For example, if you set 2000 pulse motor to rotate 1 turn, then set H05-09 parameter value 2000.

Unit conversion schematic table

Number of pulses in 1 turn	user understands the unit				Convert to units that comply with CIA402 protocol			
H05-07molecule H05-09 denominator	Target position r	Target speed r/min	Acceleration to target acceleration time	Deceleration time to 0	Target position Pul	Target speed Pul/s	Acceleratio n Pul/s ²	Deceleratio n Pul/s ²
Factory default 1000	Run to 100.5 RPM	2000 rpm per minute	2 seconds to 2000 RPM	3 seconds to slow down to a full stop	100.5*1000 =100500	2000/60*1 000 =33333	33333/2 =16666	33333/3 =11111

For example, if the control mode is PP contour position mode, as shown in the table above, the position value, speed value and acceleration and deceleration value in CIA402 are calculated according to the number of pulses in 1 turn and the target position, target speed and acceleration and deceleration time expected by the user, and then the specific steps of absolute positioning operation of the motor are controlled by SDO communication.

Step1: Use the host computer to set the driver control mode parameter H02-00 to 8 (CanOpen control mode);

Step2: Use SDO communication to write the 6060h motion mode in the CIA402 object dictionary to 1 (contour position mode);

Step3: Use SDO communication to write the contour-position pattern related object in the CIA402 object dictionary - target position 607Ah into 100500; Target speed 6081h write 33333;

Acceleration 6083h write 16666; Speed down 6084h write 11111;

Step4: Use SDO communication to write the 6040h control words in the CIA402 object dictionary into 00h→06h→07h→0Fh→1F motor to start and run, and the motor will stop automatically when it runs to 100500 position.

2.4 6040 Bit definition of the control word

Bit	0	1	2	3	4-6	7	8	9-15
Function	Enter servo and stand by	Enter the main circuit	Execute quick stop	Execute the enable run	Operation mode correlation	Fault resetting	Reserved	NA

Additional notes for other bits:

Bit 2: The quick stop trigger logic is 0 valid, **pay attention to distinguish it from other triggered logic;**

Bit 7: Slave fault reset trigger logic is rise edge effective;

Bit 4: in pp mode, the trigger logic is effective along the rise, and the update position runs;

◆ **Each bit of the control word is meaningless and must be combined with other bits to form a control instruction.**

◆ bit0~bit3 and bit7 have the same meaning in each servo mode, and commands must be sent in sequence before the servo drive can be introduced according to the CiA402 state machine switching process

Import the expected state, each command corresponds to a definite state.

◆ bit4~bit6 is related to each servo mode (please check the control instruction state transition in different modes).

Bit	0	1	2	3	4	5	6	7/8
Function	Be ready	Wait to enable	operation has been enabled	Go wrong	Be powered on	Quick stop activation	Not running	N A

Bit	9	10	11	12-13	14	15
Function	Remote control	Target reached	NA	Operation mode correlation	NA	Origin found

2.5 6041 Bit definition of the status word

Additional notes for other bits:

◆ Bit 4 will be set when the drive is powered in.

◆ Each bit of the status word is meaningless to read separately, and must be composed with other bits to feedback the current status of the servo.

◆ bit0~bit9 has the same meaning in each servo mode. After the command is sent by the control word 6040h in sequence, the servo feedback will be a certain state.

◆ bit12~bit13 is related to each servo mode (please check the control instructions under different modes).

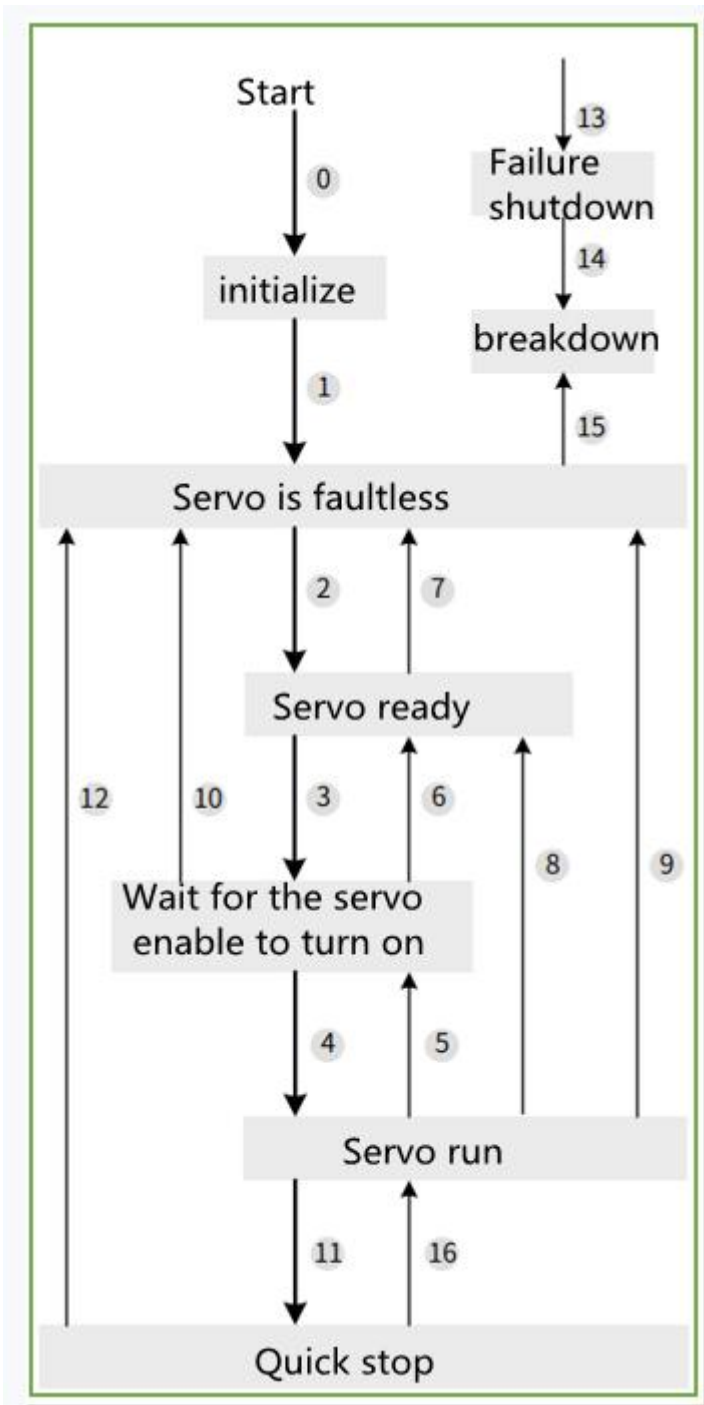
◆ bit10 and bit15 have the same meaning in each servo mode, and feedback the status of the servo after

executing a certain servo mode.

CIA402 status switch		Control word 6040	Control word 6041
Tab	Process migration	Control command	Feedback bit0-bit9
0	Power-on → Initialization	Internal automatic transition, no command required	0x0000
1	Initialization → Servo no fault	1. If a fault is detected after initialization, go to 13. 2. If there is no fault at the end of initialization, the transition will be automatic without command	0x0250
2	Servo no fault → Servo ready	0x06	0x0231
3	Servo ready → Wait to turn on the servo enable	0x07	0x0233
4	Wait for servo Enable → Servo run	0x0F	0x0237
5	Servo Run → Wait for the servo enable to turn on	0x07	0x0233
6	Wait for servo Enable → Servo ready	0x06	0x0231
7	Servo ready → Servo no fault	0x00	0x0250
8	Servo Run → Servo ready	0x06	0x0231
9	Servo run → No fault in the servo	0x00	0x0250
10	Wait to turn on the servo enable → No fault of the servo	0x00	0x0250
11	Servo operation → Quick shutdown	0x02	0x0217
12	Quick stop → Servo fault free	Automatic transition after quick shutdown, no command required	0x0250
13	→Fault shutdown	In addition to the fault state, in other states, once the driver detects an alarm, it automatically switches to the fault shutdown state without command	0x021F
14	Failure stop → Failure	Automatic transition after the shutdown is complete, no command required	0x0218
15	Fault → Servo no fault	0x80	0x0250
16	Quick stop → Servo run	When the quick stop is complete, send the command 0x0F	0x0237

2.6 Each mode controls the state transition of operation

CIA402 state transition diagram



- ◆ The user must follow the state transition table to control the CANOPEN motor;
- ◆ Each bit of the control word 6040h has no meaning and must be combined with other bits to form a control instruction.
- ◆ bit0 to bit9 of the status word 6041h has the same meaning in any servo mode, and bit10 to bit15(except bit11 and bit14) is related to the running status of each servo mode, which is represented by "0" in the table above.

2.7 6040 Control command Step

Note: To use CanOpen control, you need to use our company's debugging AIMOTOR software to set the H02-00 control mode parameter to 8 (CanOpen control).

Once set to CanOpen control must comply with CanOpen control requirements, its corresponding object dictionary takes effect; Other control modes are invalid;

Sequence control step		0	1	2	3	4	5	6	7
Mode	Control word	preparation work	initial	Ready	Wait for enable	Enable	Startup run	conjugation	Stop
CSP Cyclic synchronization position mode	6040	Establish communication, assign PDO mapping, and activate master loop synchronization	00h	06h	07h	0Fh	Master station send position send command	Master station control	Command is sent from the stop position of the master station
PP profile Position mode	6040	Establish communication status Set motion parameters related to PP mode	00h	06h	07h	0Fh	Start absolute positioning: 0Fh→1Fh		1: The position reaches the automatic stop 2: Forcibly fails to stop writing 00h/02h or 06h/07h
						4Fh	Start relative positioning: 4Fh→5Fh		
PV profile speed mode	6040	Establish the communication status and set the motion parameters related to the PV mode	00h	06h	07h	0Fh	Start running after enabling	Just change the speed	Forcible stop write 00h/02h or 06h/07h
PT profile torque mode	6040	Establish communication status and set motion parameters related to PT mode	00h	06h	07h	0Fh	Start running after enabling	Just change the torque	Forcible stop write 00h/02h or 06h/07h
HM back to origin mode	6040	Establish the communication state and set the motion parameter state related to HM mode	00h	06h	07h	0Fh	0Fh→1Fh Start	-	1: Automatically stops when the origin is found. 2: forcibly stops writing 00h/02h or 06h/07h

◆CSP cyclic synchronous position mode mode, the position is delivered by the main control unit according to the agreed synchronization cycle, the motor performs displacement according to the delivered position point and synchronous frame, and the motor running contour curve (such as position, speed, etc.) is planned and calculated by the main control unit in advance;

◆PP/PV/PT contour mode, the main control unit can use the SDO communication mode to set the necessary parameters of the mode, control the motor operation according to the mode control command steps, and the motor operation contour curve is planned and calculated by the motor according to the set parameter values (such as acceleration and deceleration speed, running speed, etc.).

2.8 6098 Origin search mode

Set value	Initial Orientation	Deceleration point	Origin	Process steps
24	Forward	Origin Switch	Origin Switch	Motor first searches the origin switch at high speed in the set direction. When it encounters the origin switch signal rising edge, it starts to run out at low speed. When it breaks away from the origin switch signal falling along the motor, it reverses and continues to search the origin switch signal rising edge at low speed.
28	Reverse	Origin Switch	Origin Switch	
34	Forward	Z Signal	Motor Z Signal	Motor first searches the Z phase at a high speed in the set direction. When it encounters the rising edge of the motor Z, it starts to reverse at a low speed. When it encounters the rising edge signal on the other side of the Z phase, it stops immediately and returns to zero successfully
33	Reverse	Z Signal	Motor Z Signal	
4	Forward	Origin Switch	Motor Z Signal	Motor first searches the origin switch at a high speed in the set direction. When it encounters the origin switch signal rising edge, it begins to reverse at a low speed and runs away from the origin switch signal falling edge. When it encounters the origin switch signal rising edge, it runs to the Z signal and stops immediately, returning to zero successfully.
6	Reverse	Origin Switch	Motor Z Signal	
18	Forward	Forward overpass	Forward overpass	Motor first searches the origin switch at a high speed in the set direction. When it encounters the rising edge of the forward overrange switch signal, it begins to reverse at a low speed. When it breaks away from the falling edge of the forward overrange switch signal, the motor reverses again to search the rising edge of the forward overrange switch signal at a low speed. When it encounters the rising edge of the forward overrange switch signal, it stops immediately and returns to zero.
17	Reverse	Reverse overpass	Reverse overpass	
2	Forward	Forward overpass	Z Signal	Motor first searches for the forward overrange switch at a high speed in the set direction. When it encounters the rising edge of the forward overrange switch signal, it begins to slow down and reverse the operation. When it breaks away from the falling edge of the forward overrange switch signal, it continues to run until the motor Z is stopped immediately and the return to zero is successful.
1	Reverse	Reverse overpass	Z Signal	
37	Forward	Mechanical limit potential	Mechanical limit potential	Motor first searches the mechanical limit position at a low speed with the set direction and torque (the torque set by H05_58). When the mechanical limit position reaches the locking and the torque reaches the upper limit of the zero torque limit and maintains the default time, the motor stops immediately and the return to zero is successful.
38	Reverse	Mechanical limit potential	Mechanical limit potential	
35	-	-	-	Set the current position as the origin

Note :

- ◆ The external switch signal useful in the origin search mode, please associate the selected corresponding DI function option, otherwise the alarm ER.601 return to zero failure will occur;

- ◆ Change the origin search mode, please modify the object parameters of the origin mode 6098h when the motor is disabled;
- ◆ In the selected mode, if the deceleration point is the origin switch, and the positive and negative overrange switch DI is associated, the motor encounters the overrange switch motor in the search path and automatically reverses to continue the search;
- ◆ In the process steps, the sub-index 1 object of the high-speed search speed 6099 index is set, and the low-speed search speed is set by the sub-index 2 of the 6099 index;
- ◆ If the origin is not found within the H05_35 parameter time, the alarm ER.601 return to zero timeout will occur;
- ◆ The DO function (OutFun16-HomeAttain) takes effect when the origin resets to zero, but takes effect when the OFF function is enabled.

2.9 607E command polarity

Bit	0	1	2	3	4	5	6	7
Funtion	NA	NA	NA	NA	NA	Torque command polarity	Speed command polarity	Speed command polarity

Bit7=1, indicating that in the PP contour position mode, the position instruction is 607Ah $\times$ (-1), and the motor is turned in reverse.

Bit6=1, indicating that in PV profile speed mode, the speed instruction is 60FFh $\times$ (-1), and the motor is turned in reverse.

Bit5=1, indicating that in PT contour torque mode, torque instruction 6071h $\times$ (-1), the motor turns in reverse.

NA: no definition